MATHEMATICS / INDEPENDENT STUDY

# MM7033

# Advanced Algebra

Exercises and mock assessments

---

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

Edition 1 · 24 October 2026

# Contents

1 Unit 1 · Fields and splitting fields ..... 3
2 Unit 2 · Existence and transcendence ..... 5
3 Unit 3 · Modules and quotient constructions ..... 7
4 Unit 4 · Free modules and exact sequences ..... 9
5 Unit 5 · Tensor products ..... 11
6 Unit 6 · Localization ..... 13
7 Unit 7 · Multilinear algebra ..... 15
8 Unit 8 · Noetherian methods and applications ..... 17
9 Mock assessment 1 ..... 19
10 Mock assessment 2 ..... 20

**How to use this volume** E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

# 1 Unit 1 · Fields and splitting fields

**MM7033-U1.01 · E · 15 min**
Learning objective: U1.1 Minimal polynomials
Find the minimal polynomial and degree of $\sqrt{11}$ over $\mathbb{Q}$.

**MM7033-U1.02 · E · 15 min**
Learning objective: U1.1 Splitting fields
Find a splitting field of $x^2 - 3$ over $\mathbb{Q}$.

**MM7033-U1.03 · E · 15 min**
Learning objective: U1.2 Finite-field subfields
List the subfield sizes in $\mathbb{F}_{3^4}$.

**MM7033-U1.04 · E · 15 min**
Learning objective: U1.2 Separability
Show $x^5 - x$ over $\mathbb{F}_5$ has no repeated roots.

**MM7033-U1.05 · C · 30 min**
Learning objective: U1.1 Splitting-field degree
Find the degree of the splitting field of $x^3 - 3$ over $\mathbb{Q}$.

**MM7033-U1.06 · C · 30 min**
Learning objective: U1.1 Embeddings
Describe all $\mathbb{Q}$-embeddings of $\mathbb{Q}\left(\sqrt{2}\right)$ into $\mathbb{C}$.

**MM7033-U1.07 · C · 30 min**
Learning objective: U1.2 Frobenius order
Prove Frobenius on $\mathbb{F}_{p^n}$ has order exactly $n$.

**MM7033-U1.08 · C · 30 min**
Learning objective: U1.2 Derivatives
Prove a polynomial over a field of characteristic $p$ has derivative zero exactly when all nonzero terms have exponents divisible by $p$.

**MM7033-U1.09 · C · 30 min**
Learning objective: U1.1 Changing the base
Compare the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ and over $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$.

**MM7033-U1.10 · A · 50 min**
Learning objective: U1.1 Quartic splitting field
Prove the splitting field of $x^4 - 2$ over $\mathbb{Q}$ has degree eight.

**MM7033-U1.11 · A · 50 min**
Learning objective: U1.2 Subfield intersections
Inside $\mathbb{F}_{p^{12}}$, determine the intersection of its subfields of orders $p^4$ and $p^6$.

**MM7033-U1.12 · A · 50 min**
Learning objective: U1.2 Inseparable irreducibles
Prove $x^p - t$ is irreducible and inseparable over $\mathbb{F}_{p(t)}$.

# 2 Unit 2 · Existence and transcendence

**MM7033-U2.01 · E · 15 min**
Learning objective: U2.1 Chains
Give a partially ordered set with two maximal elements and no greatest element.

**MM7033-U2.02 · E · 15 min**
Learning objective: U2.1 Proper ideals
Prove an ideal $I$ of a unital ring is proper exactly when $1 \notin I$.

**MM7033-U2.03 · E · 15 min**
Learning objective: U2.2 Independence
Is $\{t, t^2\}$ algebraically independent over $K$ in $K(t)$?

**MM7033-U2.04 · E · 15 min**
Learning objective: U2.2 Transcendence bases
Give a transcendence basis of an algebraic extension $\frac{L}{K}$.

**MM7033-U2.05 · C · 30 min**
Learning objective: U2.1 Chain unions
Prove the union of a nonempty chain of proper ideals is proper and an ideal.

**MM7033-U2.06 · C · 30 min**
Learning objective: U2.1 Maximal ideals over an ideal
Use maximal-ideal existence in a quotient to show every proper ideal $I$ lies in a maximal ideal.

**MM7033-U2.07 · C · 30 min**
Learning objective: U2.2 Algebraic closure
Show an algebraically closed field has no proper algebraic field extension.

**MM7033-U2.08 · C · 30 min**
Learning objective: U2.2 Transitivity
Prove algebraicity is transitive through a tower, allowing infinite extensions.

**MM7033-U2.09 · C · 30 min**
Learning objective: U2.1 Extending independent sets
Use Zorn's lemma to extend a linearly independent subset of a vector space to a basis.

**MM7033-U2.10 · A · 50 min**
Learning objective: U2.2 A nontrivial transcendence basis
Show $\{t^2\}$ is a transcendence basis of $\frac{K(t)}{K}$.

**MM7033-U2.11 · A · 50 min**
Learning objective: U2.1 Units detected by maximal ideals
Prove an element of a nonzero commutative unital ring is a unit exactly when it belongs to no maximal ideal.

**MM7033-U2.12 · A · 50 min**
Learning objective: U2.2 Maximal independence
Prove every maximal algebraically independent subset $B$ of $\frac{L}{K}$ is a transcendence basis.

# 3 Unit 3 · Modules and quotient constructions

**MM7033-U3.01 · E · 15 min**
Learning objective: U3.1 Annihilators
Compute the annihilator of $[4]$ in the $\mathbb{Z}$-module $\mathbb{Z}/12\mathbb{Z}$.

**MM7033-U3.02 · E · 15 min**
Learning objective: U3.1 Module maps
Describe all $\mathbb{Z}$-module maps $\mathbb{Z} \to \mathbb{Z}/5\mathbb{Z}$.

**MM7033-U3.03 · E · 15 min**
Learning objective: U3.2 Submodules
Show an ideal of $R$ is exactly a submodule of $R$ as an $R$-module.

**MM7033-U3.04 · E · 15 min**
Learning objective: U3.2 Quotients
Describe the quotient $\frac{\mathbb{Z}^2}{N}$ for $N = \{(a, 0) : a \in \mathbb{Z}\}$.

**MM7033-U3.05 · C · 30 min**
Learning objective: U3.1 Cyclic homomorphisms
Compute the number of $\mathbb{Z}$-module maps $\mathbb{Z}/m\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$.

**MM7033-U3.06 · C · 30 min**
Learning objective: U3.1 Operator modules
Let $T$ on $K^2$ satisfy $T(e_1) = 0, T(e_2) = e_1$. Show its $K[x]$-module is cyclic and identify it as a quotient.

**MM7033-U3.07 · C · 30 min**
Learning objective: U3.2 Isomorphism theorem
In $\mathbb{Z}$, take submodules $A = 4\mathbb{Z}$ and $B = 6\mathbb{Z}$. Identify $\frac{A+B}{B}$ and $\frac{A}{A\cap B}$.

**MM7033-U3.08 · C · 30 min**
Learning objective: U3.2 Simple modules
Classify the simple $\mathbb{Z}$-modules up to isomorphism.

**MM7033-U3.09 · C · 30 min**
Learning objective: U3.1 Torsion
For a domain $R$, prove the torsion elements of an $R$-module form a submodule.

**MM7033-U3.10 · A · 50 min**
Learning objective: U3.1 Maps from a quotient
Prove naturally that $\mathrm{Hom}_{R(\frac{R}{I},M)}$ is isomorphic to $\{m \in M : Im = 0\}$ as an $R$-module.

**MM7033-U3.11 · A · 50 min**
Learning objective: U3.2 Splitting a torsion module
Prove $\frac{\mathbb{Z}}{12}\mathbb{Z} \cong \frac{\mathbb{Z}}{3}\mathbb{Z} \oplus \frac{\mathbb{Z}}{4}\mathbb{Z}$ as modules, and identify the images of a generator.

**MM7033-U3.12 · A · 50 min**
Learning objective: U3.1 Submodule invariance
For a vector space with operator $T$, prove its $K[x]$-submodules are exactly its $T$-invariant vector subspaces.

# 4 Unit 4 · Free modules and exact sequences

**MM7033-U4.01 · E · 15 min**
Learning objective: U4.1 Presentations
Give a one-generator, one-relation presentation of $\frac{\mathbb{Z}}{9}\mathbb{Z}$.

**MM7033-U4.02 · E · 15 min**
Learning objective: U4.2 Exactness
Check exactness of $0 \to K \xrightarrow{i} K^2 \xrightarrow{p} K \to 0$, where $i(a) = (a, 0)$ and $p(a, b) = b$.

**MM7033-U4.03 · E · 15 min**
Learning objective: U4.1 Finite support
Give an element of $\prod_{n\geq 0} \mathbb{Z}$ not lying in $\bigoplus_{n\geq 0} \mathbb{Z}$.

**MM7033-U4.04 · E · 15 min**
Learning objective: U4.1 Free maps
How is a linear map $R^3 \to M$ determined?

**MM7033-U4.05 · C · 30 min**
Learning objective: U4.2 Nonsplitting
Prove $0 \to \mathbb{Z} \xrightarrow{\times 3} \mathbb{Z} \to \frac{\mathbb{Z}}{3}\mathbb{Z} \to 0$ does not split.

**MM7033-U4.06 · C · 30 min**
Learning objective: U4.2 Free quotients
Show a short exact sequence with quotient $R^n$ splits.

**MM7033-U4.07 · C · 30 min**
Learning objective: U4.1 Presentation computation
Identify the $\mathbb{Z}$-module generated by $u, v$ with relations $u - 2v = 0$ and $6v = 0$.

**MM7033-U4.08 · C · 30 min**
Learning objective: U4.2 Direct summands
Prove a direct summand of a projective module is projective.

**MM7033-U4.09 · C · 30 min**
Learning objective: U4.1 Finite rank
Prove $R^2$ cannot be isomorphic to $R^3$ for a nonzero commutative unital ring.

**MM7033-U4.10 · A · 50 min**
Learning objective: U4.2 Projective cyclic example
Show $\frac{\mathbb{Z}}{2}\mathbb{Z}$ is not a projective $\mathbb{Z}$-module.

**MM7033-U4.11 · A · 50 min**
Learning objective: U4.2 A projective nonfree module
Let $R = K \times K$ and $e = (1, 0)$. Prove $\mathfrak{R}$ is projective but is not free.

**MM7033-U4.12 · A · 50 min**
Learning objective: U4.2 Splitting from a retraction
Given a short exact sequence and a retraction $r : B \to A$ of its injection $i$, construct the isomorphism $B \cong A \oplus C$.

# 5 Unit 5 · Tensor products

**MM7033-U5.01 · E · 15 min**
Learning objective: U5.1 Tensor relations
Simplify $(2m) \otimes n + m \otimes (3n)$ over $\mathbb{Z}$.

**MM7033-U5.02 · E · 15 min**
Learning objective: U5.2 Cyclic tensors
Compute $\frac{\mathbb{Z}}{8}\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{12}\mathbb{Z}$.

**MM7033-U5.03 · E · 15 min**
Learning objective: U5.1 Unit tensor
Give inverse maps proving $R \otimes_R M \cong M$.

**MM7033-U5.04 · E · 15 min**
Learning objective: U5.2 Tensor annihilation
Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{7}\mathbb{Z}$.

**MM7033-U5.05 · C · 30 min**
Learning objective: U5.1 Pure tensors
Show $e_1 \otimes e_1 + e_2 \otimes e_2$ in $K^2 \otimes K^2$ is not a pure tensor.

**MM7033-U5.06 · C · 30 min**
Learning objective: U5.1 Tensor basis
Prove $R^m \otimes_R R^n$ is free of rank $mn$.

**MM7033-U5.07 · C · 30 min**
Learning objective: U5.2 Quotient tensors
Compute $\frac{R}{I} \otimes_R \frac{R}{J}$.

**MM7033-U5.08 · C · 30 min**
Learning objective: U5.2 Failure of injectivity
Tensor $0 \to \mathbb{Z} \xrightarrow{\times 4} \mathbb{Z} \to \frac{\mathbb{Z}}{4}\mathbb{Z} \to 0$ with $\frac{\mathbb{Z}}{2}\mathbb{Z}$. Describe every resulting map.

**MM7033-U5.09 · C · 30 min**
Learning objective: U5.1 Distributivity
Prove $(M \oplus N) \otimes P \cong (M \otimes P) \oplus (N \otimes P)$.

**MM7033-U5.10 · A · 50 min**
Learning objective: U5.2 Right exactness
For a submodule $K \leq M$, identify the kernel of $M \otimes N \to \left(\frac{M}{K}\right) \otimes N$.

**MM7033-U5.11 · A · 50 min**
Learning objective: U5.1 Tensor–Hom correspondence
Prove $\mathrm{Hom}_{R(M \otimes N, P)} \cong \mathrm{Hom}_{R\left(M, \mathrm{Hom}_{R(N,P)}\right)}$ naturally.

**MM7033-U5.12 · A · 50 min**
Learning objective: U5.2 A torsion tensor vanishes
Prove $\left(\frac{\mathbb{Q}}{\mathbb{Z}}\right) \otimes_{\mathbb{Z}\left(\frac{\mathbb{Q}}{\mathbb{Z}}\right)} = 0$.

# 6 Unit 6 · Localization

**MM7033-U6.01 · E · 15 min**
Learning objective: U6.1 Localized integers
Describe $\mathbb{Z}\left[\frac{1}{3}\right]$ as a subring of $\mathbb{Q}$.

**MM7033-U6.02 · E · 15 min**
Learning objective: U6.1 Units at a prime
Which of $2, 5, 10$ are units in $\mathbb{Z}_{(5)}$?

**MM7033-U6.03 · E · 15 min**
Learning objective: U6.1 Vanishing
In $\frac{\mathbb{Z}}{10}\mathbb{Z}$, localize at powers of two. Show the class of five becomes zero.

**MM7033-U6.04 · E · 15 min**
Learning objective: U6.2 Localizing a cyclic module
Compute $\mathbb{Z}\left[\frac{1}{2}\right] \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{8}\mathbb{Z}$.

**MM7033-U6.05 · C · 30 min**
Learning objective: U6.1 Injectivity criterion
Prove the map $R \to S^{-1}R$ is injective exactly when no element of $S$ annihilates a nonzero element of $R$.

**MM7033-U6.06 · C · 30 min**
Learning objective: U6.1 Identifying a localization
Prove localizing $\frac{\mathbb{Z}}{6}\mathbb{Z}$ at powers of two gives $\mathbb{F}_3$.

**MM7033-U6.07 · C · 30 min**
Learning objective: U6.2 A common annihilator
If $M$ is finitely generated and $S^{-1}M = 0$, prove some $s \in S$ kills all of $M$.

**MM7033-U6.08 · C · 30 min**
Learning objective: U6.2 Quotients and localization
Show $S^{-1}\left(\frac{M}{N}\right) \cong S^{-1}\frac{M}{\operatorname{im}(S^{-1}N)}$.

**MM7033-U6.09 · C · 30 min**
Learning objective: U6.1 Local rings
Show the nonunits in $R_{\mathfrak{p}}$ form its unique maximal ideal.

**MM7033-U6.10 · A · 50 min**
Learning objective: U6.2 Detecting zero locally
Prove an $R$-module $M$ is zero if $M_{\mathfrak{m}} = 0$ for every maximal ideal $\mathfrak{m}$. No finite-generation hypothesis is allowed.

**MM7033-U6.11 · A · 50 min**
Learning objective: U6.1 Localization in stages
Show inverting $f$ and then $g$ in a commutative ring is equivalent to inverting $fg$.

**MM7033-U6.12 · A · 50 min**
Learning objective: U6.2 Flatness of localization
Prove $S^{-1}R$ is flat by explicitly checking that an injection remains injective after localization and identifying localization with tensoring.

# 7 Unit 7 · Multilinear algebra

**MM7033-U7.01 · E · 15 min**
Learning objective: U7.1 Exterior arithmetic
Expand $(e_1 + 2e_2) \wedge (3e_1 + e_2)$ in $\wedge^2 R^2$.

**MM7033-U7.02 · E · 15 min**
Learning objective: U7.1 Exterior rank
Give a basis and rank of $\wedge^2 R^4$.

**MM7033-U7.03 · E · 15 min**
Learning objective: U7.2 Trace and determinant
Find trace and determinant of $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ over $\mathbb{Z}$.

**MM7033-U7.04 · E · 15 min**
Learning objective: U7.1 Repeated inputs
Why is $v \wedge v = 0$ even in characteristic two?

**MM7033-U7.05 · C · 30 min**
Learning objective: U7.2 Cyclic trace
Prove $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for square matrices over a commutative ring.

**MM7033-U7.06 · C · 30 min**
Learning objective: U7.2 Similarity invariance
Prove similar matrices have equal determinant and trace.

**MM7033-U7.07 · C · 30 min**
Learning objective: U7.1 Functorial exterior powers
Show $\wedge^{n(TS)} = (\wedge^n T)(\wedge^n S)$.

**MM7033-U7.08 · C · 30 min**
Learning objective: U7.2 Rank-one endomorphisms
Let $T : M \to M$ have form $T(m) = \varphi(m)v$ on a finite free module. Prove $\operatorname{tr}(T) = \varphi(v)$.

**MM7033-U7.09 · C · 30 min**
Learning objective: U7.1 Vanishing top powers
Prove $\wedge^n R^r = 0$ for $n > r$.

**MM7033-U7.10 · A · 50 min**
Learning objective: U7.2 Dual bases
For a finite projective module, prove the trace formula is independent of the chosen dual basis.

**MM7033-U7.11 · A · 50 min**
Learning objective: U7.1 Exterior direct-sum decomposition
For finite free modules $M, N$, prove $\wedge^2 (M \oplus N) \cong \wedge^2 M \oplus (M \otimes N) \oplus \wedge^2 N$.

**MM7033-U7.12 · A · 50 min**
Learning objective: U7.2 Finite-order operators
Let $T^5 = 1$ on a finite-dimensional vector space. Show $\det(T)^5 = 1$, and give a reason this does not force $T$ to be the identity.

# 8 Unit 8 · Noetherian methods and applications

**MM7033-U8.01 · E · 15 min**
Learning objective: U8.1 Noetherian examples
Explain why every finite-dimensional vector space is a Noetherian module over its field.

**MM7033-U8.02 · E · 15 min**
Learning objective: U8.1 Hilbert's theorem
Why is $K[x, y, z]$ Noetherian for a field $K$?

**MM7033-U8.03 · E · 15 min**
Learning objective: U8.2 Coordinate rings
Show $\frac{K[x,y]}{xy}$ is not a domain.

**MM7033-U8.04 · E · 15 min**
Learning objective: U8.2 Specialization
Compute $R^3 \otimes_R \frac{R}{M}$ for a maximal ideal $M$.

**MM7033-U8.05 · C · 30 min**
Learning objective: U8.1 Quotients
Prove a quotient of a Noetherian module is Noetherian.

**MM7033-U8.06 · C · 30 min**
Learning objective: U8.1 Finitely generated submodules
Prove the ascending chain condition implies every submodule is finitely generated.

**MM7033-U8.07 · C · 30 min**
Learning objective: U8.1 Infinite polynomial ring
Show $K[x_1, x_2, \ldots]$ is not Noetherian.

**MM7033-U8.08 · C · 30 min**
Learning objective: U8.2 Local coordinate calculation
Compute the localization at $x$ of $A = \frac{K[x,y]}{xy}$.

**MM7033-U8.09 · C · 30 min**
Learning objective: U8.2 Finite presentation
Prove every finitely generated module over a Noetherian ring is finitely presented.

**MM7033-U8.10 · A · 50 min**
Learning objective: U8.1 Localization preserves Noetherianity
Prove a localization of a Noetherian ring is Noetherian.

**MM7033-U8.11 · A · 50 min**
Learning objective: U8.2 Eigenspace decomposition
Suppose $T^2 = 1$ on a vector space over a field of characteristic not two. Construct projections onto the two eigenspaces and prove a direct-sum decomposition.

**MM7033-U8.12 · A · 50 min**
Learning objective: U8.1 Hilbert reduction step
In the proof of Hilbert's theorem, explain in detail why finitely many polynomials with leading coefficients generating the leading-coefficient ideal suffice together with finitely many bounded-degree polynomials.

# 9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

**MM7033-M1.01 · E · 15 min**
Learning objective: U1 Finite fields
List the subfield sizes of $\mathbb{F}_{2^8}$ and justify that the list is complete.

**MM7033-M1.02 · E · 15 min**
Learning objective: U5 Tensor computation
Compute $\frac{\mathbb{Z}}{15}\mathbb{Z} \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{21}\mathbb{Z}$.

**MM7033-M1.03 · C · 30 min**
Learning objective: U3–U4 Presentations
Identify the module generated by $a, b$ with relations $a = 3b$ and $10b = 0$, and decide whether it is projective over $\mathbb{Z}$.

**MM7033-M1.04 · C · 30 min**
Learning objective: U6 Localization
Compute $\mathbb{Z}\left[\frac{1}{3}\right] \otimes_{\mathbb{Z}} \frac{\mathbb{Z}}{12}\mathbb{Z}$.

**MM7033-M1.05 · A · 50 min**
Learning objective: U2–U3 Simple quotients
Prove every nonzero finitely generated module over a commutative unital ring has a maximal proper submodule.

**MM7033-M1.06 · A · 50 min**
Learning objective: U7–U8 Operators
Let $T^3 = 1$ over $\mathbb{C}$. Prove the space is a direct sum of the three eigenspaces and describe projections using interpolation.

# 10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

**MM7033-M2.01 · E · 15 min**
Learning objective: U3 Module maps
Compute $\mathrm{Hom}_{\mathbb{Z}(\frac{\mathbb{Z}}{6}\mathbb{Z},\frac{\mathbb{Z}}{9}\mathbb{Z})}$ as an abelian group.

**MM7033-M2.02 · E · 15 min**
Learning objective: U7 Exterior powers
Compute the rank of $\wedge^3 R^5$ and explain why $\wedge^6 R^5 = 0$.

**MM7033-M2.03 · C · 30 min**
Learning objective: U1 Splitting fields
Find the degree of the splitting field of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

**MM7033-M2.04 · C · 30 min**
Learning objective: U8 Noetherianity
Prove $\frac{K[x,y]}{x^2,xy}$ is Noetherian. Must it be a domain?

**MM7033-M2.05 · A · 50 min**
Learning objective: U5–U6 Local detection of a map
Prove a module map is an isomorphism if its localization at every maximal ideal is an isomorphism.

**MM7033-M2.06 · A · 50 min**
Learning objective: U4–U7 Projective trace
Let $M$ be finite projective and $T(m) = \varphi(m)v$. Prove $\mathrm{tr}(T) = \varphi(v)$ using a dual basis.